

INDIAN E-COMMERCE

Sales & Profitability Analysis

Complete Data Analytics Project Report

Python • Pandas • MySQL • SQL • Views • Stored Procedures • Power BI • DAX

Project Snapshot	Result
Revenue analyzed	■431,502
Profit analyzed	■23,955
Overall profit margin	5.55%
Units sold	5,615
Unique customers	332
Analysis period	April 2018 – March 2019

Project purpose: Convert raw Indian e-commerce data into reliable business insights and an interactive Power BI dashboard, with a strong focus on revenue, profitability, customers, geography, products, and target achievement.

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Note: This report documents the work completed in the project, including the corrections made during analysis and dashboard development. Where a business cause is not directly present in the dataset, the report avoids presenting it as a confirmed fact.

1. Executive Summary

This project is an end-to-end Indian e-commerce sales and profitability analysis. The work began with raw CSV data, moved through Python-based cleaning and exploratory analysis, then into MySQL for structured SQL analysis, reusable views and selected stored procedures, and finally into Power BI for interactive reporting and DAX-based KPIs.

The central analytical lesson from the project is that revenue alone is not sufficient for business decision-making. Several parts of the data demonstrate that a category, product, customer or state can generate substantial revenue while producing comparatively low or even negative profit.

Area	Main Result
Overall Revenue	■431,502
Overall Profit	■23,955
Profit Margin	5.55%
Units Sold	5,615
Customers	332
Revenue Leader	Electronics
Profit Leader	Clothing
Lowest Category Margin	Furniture – 1.81%
Worst Profit Month	June 2018 – -■4,970
Largest Loss Sub-Category	Tables – -■4,011
Best Overall Target Achievement	Electronics – 128.11%

Executive takeaway: Electronics is strongest for revenue and target achievement; Clothing is strongest for total profit and margin; Furniture needs profitability attention; Tables are a major loss area; and repeat customers show substantially higher value per customer than one-time customers in the analyzed data.

2. Business Problem & Objectives

Business Problem

The business needs a clear view of where sales are coming from, where profit is being generated or lost, which customers are valuable, how performance changes over time, and whether actual sales are meeting planned targets.

Project Objectives

- Clean and validate the raw datasets before analysis.
- Calculate revenue, profit, profit margin, units sold and customer counts.
- Compare performance across categories and sub-categories.
- Identify loss-making products and transactions.
- Analyze state-level revenue, profit and profit margin.
- Study monthly revenue and profit trends.
- Compare repeat and one-time customers.
- Compare actual sales against category and monthly targets.
- Build reusable SQL views and selected parameterized procedures.
- Build an interactive Power BI dashboard with DAX measures.

Analytical Approach

The project follows a practical business-analysis sequence: **Raw Data** → **Cleaning** → **Validation** → **EDA** → **MySQL** → **SQL Analysis** → **Views/Procedures** → **Data Model** → **DAX** → **Power BI Dashboard** → **Business Insights**.

3. Dataset & Data Structure

List of Orders

Column	Role
Order ID	Order identifier
Order Date	Date of order
CustomerName	Customer name
State	Customer/order state
City	Customer/order city

Order Details

Column	Role
Order ID	Join key to order information
Amount	Sales/revenue amount
Profit	Profit amount
Quantity	Units sold
Category	Product category
Sub-Category	Product sub-category

Sales Target

Column	Role
Month of Order Date	Target month
Category	Target category
Target	Planned sales target

Important grain consideration: Order Details is at line-item level, while Sales Target is at month + category level. Directly joining target rows to every transaction detail can duplicate targets. The project corrected this by aggregating actual sales to month + category before comparing them to target.

4. Python Data Cleaning

Removing Completely Blank Rows

The List of Orders dataset originally contained 560 rows. The final 60 rows were completely blank. They were removed using Pandas:

```
list_order = list_order.dropna(how='all')
```

The resulting List of Orders table contained 500 valid rows.

Validation of Missing Values and Duplicates

- List of Orders: no missing values after blank-row removal.
- Order Details: 1,500 rows, no missing values and no exact duplicate rows.
- Sales Target: 36 records, no missing values.
- Repeated Order IDs in Order Details were retained because one order can contain multiple line items.

Date Conversion

```
list_order['Order Date'] = pd.to_datetime(  
    list_order['Order Date'],  
    format='%d-%m-%Y'
```

```
)

sales_target['Month of Order Date'] = pd.to_datetime(
    sales_target['Month of Order Date'],
    format='%b-%y'
)
```

The Order Date source used DD-MM-YYYY format, while the target month used abbreviated month-year values such as Apr-18. Converting them to proper dates made monthly analysis reliable.

5. Exploratory Data Analysis

Overall Customer Counts

The analysis identified 332 unique customers. A customer-frequency check found 103 repeat customers and 229 one-time customers.

Overall Financial Metrics

Total revenue was ₹431,502 and total profit was ₹23,955. The resulting overall profit margin was 5.55%. Total units sold were 5,615.

Why Revenue and Profit Were Separated

A major objective was to avoid treating revenue as the same thing as business success. For example, Electronics had the highest revenue, but Clothing produced more total profit. Similarly, Tables generated ₹22,614 revenue but produced -₹4,011 profit.

Important Calculation Definitions

Metric	Definition used
Revenue	Sum of Amount
Profit	Sum of Profit
Profit Margin	$\text{Profit} \div \text{Revenue} \times 100$
Units Sold	Sum of Quantity
Profit per Unit	$\text{Profit} \div \text{Quantity}$
Target Achievement	$\text{Actual Revenue} \div \text{Target}$

6. SQL Database & Business Analysis

Loading the Cleaned Data into MySQL

The cleaned Pandas DataFrames were uploaded into the E_Commerce MySQL database using SQLAlchemy. The main tables were created as list_order, orders, and sales_target.

```
list_order.to_sql("list_order", engine, if_exists="replace", index=False)
orders.to_sql("orders", engine, if_exists="replace", index=False)
sales_target.to_sql("sales_target", engine, if_exists="replace", index=False)
```

Core SQL Techniques Used

- SELECT and WHERE for filtering.
- GROUP BY and aggregate functions for business summaries.
- HAVING for filtering grouped/aggregate results.
- JOIN for connecting order and detail information.
- COUNT(DISTINCT) for unique customer/order counts.
- CASE for conditional calculations.
- ROUND and DATE_FORMAT for presentation and time analysis.

- CTEs for controlling aggregation grain in target analysis.

Example: Category Performance

```
SELECT
    Category,
    SUM(Amount) AS total_revenue,
    SUM(Profit) AS total_profit,
    SUM(Quantity) AS total_quantity
FROM orders
GROUP BY Category
ORDER BY total_revenue DESC;
```

Important SQL Learning

WHERE vs HAVING: WHERE filters individual rows before grouping. HAVING filters the grouped result after aggregate calculations. This distinction was important when identifying repeat customers and other aggregate conditions.

7. Category & Sub-Category Profitability

Category Performance

Category	Revenue	Profit	Margin
Electronics	■165,267	■10,494	6.35%
Clothing	■139,054	■11,163	8.03%
Furniture	■127,181	■2,298	1.81%

Key finding: Electronics is the revenue leader, while Clothing is the profit and margin leader. Furniture's 1.81% margin is comparatively weak.

Top Profitable Sub-Categories

Sub-Category	Revenue	Profit	Profit/Unit
Printers	■58,252	■5,964	■20.49
Bookcases	■56,861	■4,888	■16.46
Accessories	■21,728	■3,559	■13.58
Trousers	■30,039	■2,847	■21.09
Stole	■18,546	■2,559	■3.81

Loss-Making Sub-Categories

Sub-Category	Revenue	Profit
Tables	■22,614	-■4,011
Electronic Games	■39,168	-■1,236

The loss analysis was extended to individual transactions. The ten worst individual transactions by profit included losses from Tables, Electronic Games, Phones and other sub-categories, showing that losses were not isolated to a single product family.

Important analytical lesson: High quantity is not the same as high profit. Saree sold 782 units but generated only ■352 profit, while Printers sold 291 units and generated ■5,964 profit.

8. Geographical & Monthly Analysis

Top States by Revenue

State	Revenue
Madhya Pradesh	■105,140

State	Revenue
Maharashtra	■95,348
Delhi	■22,531
Uttar Pradesh	■22,359
Rajasthan	■21,149

Top States by Profit

State	Profit
Maharashtra	■6,176
Madhya Pradesh	■5,551
Uttar Pradesh	■3,237
Delhi	■2,987
West Bengal	■2,500

Madhya Pradesh led revenue, while Maharashtra led absolute profit. The highest observed state-level profit margin was West Bengal at 17.75%.

Monthly Performance

Month	Revenue	Profit
Apr 2018	■32,726	-■3,960
May 2018	■28,545	-■3,584
Jun 2018	■23,658	-■4,970
Jul 2018	■12,966	-■2,138
Aug 2018	■30,899	-■2,180
Sep 2018	■26,628	-■4,963
Oct 2018	■31,615	■3,093
Nov 2018	■48,086	■11,619
Dec 2018	■37,579	■5,284
Jan 2019	■61,439	■9,760
Feb 2019	■38,424	■5,917
Mar 2019	■58,937	■10,077

Worst profit month: June 2018, with a loss of ■4,970. The project drilled into June by category and sub-category to understand where the loss was concentrated.

9. Customer Analysis

Top Customers by Revenue

Customer	Revenue	Profit
Yaanvi	■9,177	■488
Pooja	■9,030	-■269
Abhishek	■8,135	■1,314
Surabhi	■6,889	■412
Soumya	■6,869	■894
Harshal	■6,026	■864

Customer	Revenue	Profit
Priyanka	■5,762	■1,340
Shruti	■5,750	■485
Abhijeet	■5,691	■1,562
Sarita	■5,449	■1,265

Loss-Making Customers

Customer	Revenue	Profit
Shishu	■3,803	-■1,836
Aarushi	■4,701	-■1,669
Bharat	■1,579	-■1,153
Savi	■1,361	-■980
Sauptik	■3,287	-■927

Customer Drill-Down Example: Shishu

Shishu's largest loss was associated with a Bookcases transaction: Order B-25798, dated 2018-10-01, with ■2,830 revenue, -■1,981 profit and 13 units.

A broader Bookcases analysis showed that Bookcases was profitable overall (■56,861 revenue and ■4,888 profit), so one customer's loss did not represent the performance of the entire sub-category.

Drill-down methodology: Customer → Sub-Category → Order/Transaction. This approach helps distinguish an isolated transaction problem from a broader product-level issue.

10. Repeat vs One-Time Customers

Customer Classification

Customers were classified using the number of distinct Order IDs associated with each CustomerName. Customers with more than one distinct order were classified as Repeat; the others as One-time.

```
Customer Type =
VAR OrderCount =
    CALCULATE(
        DISTINCTCOUNT('e_commerce list_order'[Order ID]),
        ALLEXCEPT(
            'e_commerce list_order',
            'e_commerce list_order'[CustomerName]
        )
    )
RETURN
    IF(OrderCount > 1, "Repeat", "One-time")
```

Results

Customer Type	Customers	Revenue	Avg Revenue/Customer	Avg Profit/Customer
One-time	229	■191,650	■836.90	■29.03
Repeat	103	■239,852	■2,328.66	■168.02

Key insight: Repeat customers generated about 2.78× more revenue per customer and 5.79× more profit per customer than one-time customers in this dataset. This is an observed association, not a causal claim.

Recommended Dashboard Visual

Use Customer Type as the donut/legend category and Revenue or Profit as the value. The most useful supporting slicer for this analysis is Customer Type itself, allowing the dashboard user to isolate Repeat or One-time customers.

11. Sales Target vs Actual Analysis

Why the First Target Calculation Was Incorrect

The target table is stored at month + category grain, while order details contain multiple transaction rows. A direct aggregation after joining these tables can repeat the same target value for every detail row and produce incorrect achievement percentages.

Correct Approach

Actual revenue was first aggregated by month and category. That aggregated result was then joined to the target table using month and category. This aligned both datasets at the same analytical grain.

```
WITH actual_sales AS (
  SELECT
    DATE_FORMAT(l.`Order Date`, '%Y-%m') AS month,
    o.Category,
    SUM(o.Amount) AS actual_revenue
  FROM list_order l
  JOIN orders o
    ON l.`Order ID` = o.`Order ID`
  GROUP BY month, o.Category
)
SELECT
  a.Category,
  SUM(a.actual_revenue) AS actual_revenue,
  SUM(s.Target) AS total_target,
  ROUND(
    SUM(a.actual_revenue) / SUM(s.Target) * 100, 2
  ) AS achievement_pct
FROM actual_sales a
JOIN target s
  ON a.month = DATE_FORMAT(s.`Month of Order Date`, '%Y-%m')
  AND a.Category = s.Category
GROUP BY a.Category
ORDER BY achievement_pct DESC;
```

Overall Target Results

Category	Actual	Target	Achievement
Electronics	■165,267	■129,000	128.11%
Furniture	■127,181	■132,900	95.70%
Clothing	■139,054	■174,000	79.92%

Electronics exceeded target by approximately ■36.27K. Furniture missed by approximately ■5.72K. Clothing had the largest overall target shortfall at approximately ■34.95K.

The strongest monthly achievement observed was Electronics in December at 206.22%. The weakest was Clothing in July at 21.29%, a shortfall of approximately ■11.02K.

12. SQL Views & Stored Procedures

Reusable SQL Views

The project created reusable views for common business analysis:

- category_performance
- state_performance
- monthly_performance
- customer_performance
- target_vs_actual

Why Views Were Useful

A view stores a reusable query definition so the same business logic can be accessed repeatedly with SELECT statements. This is appropriate for recurring dashboard and analytical datasets.

Stored Procedures

Stored procedures were used selectively for parameterized analysis, including sales by state and sales by date range. The project intentionally avoided creating a procedure for every simple KPI because simple aggregations are more naturally handled by queries, measures, or views.

```
CALL GetSalesByState('Maharashtra');

CALL GetSalesByDate(
    '2018-04-01',
    '2018-06-30'
);
```

SQL design lesson: Use a view when you need a reusable result set/query, and a stored procedure when reusable logic benefits from parameters or procedural steps.

13. Power BI Data Model

Tables Used

- e_commerce list_order
- e_commerce orders
- e_commerce target
- MonthTable
- CategoryTable
- _measures (independent measures table)

Relationships

The model uses the order relationship from List of Orders to Order Details, plus bridge dimensions for month and category so target values remain at their correct grain.

```
MonthTable → e_commerce list_order
MonthTable → e_commerce target
```

```
CategoryTable → e_commerce orders
CategoryTable → e_commerce target
```

```
e_commerce list_order
    1
    ↓
    Order ID
    ↓
    *
e_commerce orders
```

MonthTable

```
MonthTable =
DISTINCT(
    UNION(
        SELECTCOLUMNS(
            'e_commerce list_order',
            "MonthStart",
            'e_commerce list_order'[MonthStart]
        ),
        SELECTCOLUMNS(
            'e_commerce target',
            "MonthStart",
            'e_commerce target'[Month of Order Date]
        )
    )
)
```

CategoryTable

```
CategoryTable =
DISTINCT(
    'e_commerce target'[Category]
)
```

The _measures table is intentionally independent and contains measures rather than relationships.

14. DAX Measures

Core KPI Measures

```
Total Revenue =  
SUM('e_commerce orders'[Amount])  
  
Total Profit =  
SUM('e_commerce orders'[Profit])  
  
Total Units Sold =  
SUM('e_commerce orders'[Quantity])  
  
Total Customers =  
DISTINCTCOUNT('e_commerce list_order'[CustomerName])
```

Profitability Measures

```
Profit Margin % =  
DIVIDE(  
    [Total Profit],  
    [Total Revenue],  
    0  
)  
  
Actual Revenue =  
SUM('e_commerce orders'[Amount])  
  
Sales Target =  
SUM('e_commerce target'[Target])  
  
Target Achievement % =  
DIVIDE(  
    [Actual Revenue],  
    [Sales Target],  
    0  
)  
  
Target Variance =  
[Actual Revenue] - [Sales Target]
```

For Profit Margin % and Target Achievement %, the recommended Power BI practice is to format the measure as Percentage and not multiply by 100 in the DAX expression.

Why DIVIDE Was Used

DIVIDE safely handles a zero or invalid denominator through its alternate result argument. It is preferable to a direct division operator for these dashboard measures.

15. Dashboard Design & Visuals

Dashboard Pages

Page	Purpose	Main Visuals
Executive Sales Overview	Overall business performance	KPI cards, revenue by category, monthly revenue/profit, top states
Profitability & Customer Analysis	Profit and customer behavior	Sub-category profit, margins, customers, repeat vs one-time
Target & Performance	Plan vs actual performance	Actual vs target, achievement %, variance, monthly/category analysis

Recommended Slicers

- Date – Order Date
- State
- Category
- Sub-Category
- Customer Type

The dashboard should avoid excessive slicers. Customer Name is better kept out of the main dashboard unless a detailed customer-analysis page specifically requires it.

Visual Story

The dashboard was designed to move from high-level KPIs to business explanation: overall financial performance → category performance → monthly trend → geography → sub-category profitability → customer behavior → target achievement → key insights.

16. Dashboard Screenshot & Interpretation

The following screenshot shows the dashboard version developed during the project. It combines KPI cards, category analysis, monthly trend, state performance, sub-category profitability, target comparison, customer analysis and slicers in a single executive-style layout.



Visible Dashboard Components

- Revenue, Profit, Quantity, Profit Margin and Total Customer KPI cards.
- Revenue by Category donut chart.
- Revenue & Profit by Month trend chart.
- Top Products / Sub-Categories chart.
- Actual Revenue vs Target Revenue.
- Top 5 States by Profit map.
- Repeat vs One-Time Customer Revenue & Profit.
- Top 10 Customers by Revenue.
- Category and State filtering controls.

Design observation: The dashboard emphasizes business readability by grouping financial KPIs at the top and detailed diagnostic visuals underneath. The dark theme and strong contrast support an executive-style presentation.

17. Key Business Findings

Finding Area	Conclusion
Revenue vs Profit	Electronics leads revenue, but Clothing leads total profit and profit margin.
Furniture	Furniture has ₹127.18K revenue but only ₹2.30K profit, producing a 1.81% margin.
Tables	Tables are the largest negative sub-category at -₹4.01K profit.
Electronic Games	Electronic Games also has negative overall profit at -₹1.24K.
Geography	Madhya Pradesh leads revenue; Maharashtra leads absolute profit.
Monthly	June 2018 is the worst profit month at -₹4,970.
Targets	Electronics exceeds its overall target at 128.11%; Clothing is lowest at 79.92%.
Customers	Repeat customers have much higher average revenue and profit per customer.
Volume vs Profit	Saree demonstrates that high units sold do not automatically create high profit.
Loss concentration	Electronic Games has more loss-making orders, while Tables have a larger overall negative profit.

18. Business Recommendations

- **Review Furniture profitability:** investigate why a ₹127.18K revenue category produces only ₹2.30K profit.
- **Investigate Tables:** this sub-category has the largest negative overall profit and deserves transaction-level review.
- **Review Electronic Games:** the sub-category has 40 loss-making orders and negative overall profit.
- **Strengthen repeat-customer retention:** repeat customers show substantially higher average revenue and profit per customer in the analyzed data.
- **Improve Clothing target achievement:** Clothing has the largest overall target shortfall, with especially weak performance in July.
- **Investigate June 2018:** drill from month to category and sub-category before making pricing or product decisions.
- **Monitor profitability, not just sales:** dashboard KPIs should always show revenue together with profit and margin.
- **Use target analysis at the correct grain:** maintain month + category alignment to avoid duplicated target values.

These recommendations are based on observed patterns in the dataset. The data does not contain enough information to confirm specific causes such as discounts, supplier costs, or pricing strategy, so such causes should be investigated separately rather than assumed.

19. Project Challenges & Lessons Learned

Challenge 1: Blank Rows

The initial List of Orders row count appeared larger because the dataset contained completely blank rows. Removing rows where all values were missing corrected the dataset to 500 valid rows.

Challenge 2: Duplicate Order IDs

Repeated Order IDs in Order Details were initially concerning, but they were correctly understood as multiple line items belonging to the same order. Exact duplicate-row checks confirmed there were no identical duplicate records.

Challenge 3: Target Duplication

A first target calculation produced unrealistic achievement percentages because target values were repeated when joined at transaction grain. The solution was to aggregate actuals first at month + category level.

Challenge 4: Power BI Relationships

The Power BI model required bridge dimensions for Month and Category. This prevented direct target duplication and allowed slicers and target visuals to work correctly.

Challenge 5: Customer Type Visual

Customer Type was created as a calculated column because the dashboard needs Repeat and One-time as categorical values for legends and slicing. A measure alone is not ideal for that categorical role.

Major Learning

The strongest lesson from the project is **grain awareness**: before joining two tables, understand what one row represents in each table. Many SQL and Power BI errors come from mixing different levels of detail.

20. Interview / Portfolio Summary

Short Interview Explanation

"I built an end-to-end Indian e-commerce analytics project using Python, MySQL, SQL and Power BI. I cleaned and validated three datasets, performed exploratory analysis, created SQL views and parameterized stored procedures, designed a Power BI data model with bridge dimensions for month and category, and created DAX measures for revenue, profit, margin, customers and target achievement. The analysis found that Electronics leads revenue and target achievement, Clothing leads profit and margin, Furniture has weak profitability, Tables are a major loss area, and repeat customers generate much higher value per customer."

Resume-Ready Project Description

Indian E-Commerce Sales & Profitability Analysis | Python, SQL, MySQL, Power BI, DAX

Cleaned and analyzed Indian e-commerce order data; performed SQL-based category, customer, product, state, monthly and target analysis; built reusable views and parameterized procedures; developed a Power BI model and interactive dashboard with KPI cards, trend analysis, customer segmentation, profitability analysis and target-vs-actual reporting.

Skills Demonstrated

- Python / Pandas data cleaning
- Exploratory data analysis
- MySQL database handling
- SQL joins and aggregations
- CTEs and grain-aware analysis
- SQL views and stored procedures
- Power BI data modeling
- DAX measures
- Business KPI design
- Dashboard storytelling
- Root-cause drill-down analysis

21. Conclusion

This project demonstrates a complete analytics workflow from raw data to business decision support. The project did not stop at calculating total sales; it investigated where revenue came from, where profit was generated, where losses occurred, which customers were valuable, how states and months performed, and whether sales targets were achieved.

The combination of Python, SQL, MySQL, Power BI and DAX created a strong end-to-end portfolio project. The most important analytical findings are that Electronics is the strongest revenue and target performer, Clothing is the strongest category for profit and margin, Furniture requires profitability attention, Tables are a major loss area, June 2018 was the weakest profit month, and repeat customers show substantially higher average value than one-time customers.

Final Project Takeaway: A good business dashboard should not simply show what happened. It should help explain **where performance is strong, where performance is weak, why further investigation is needed, and what the business should monitor next.**